

SYSTEMS THINKING

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



Outline

- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification

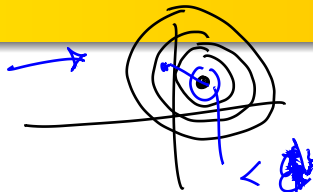


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1 Introduction to Systems Thinking

2 Systems Properties

3 Systems Classification



Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.

thinking

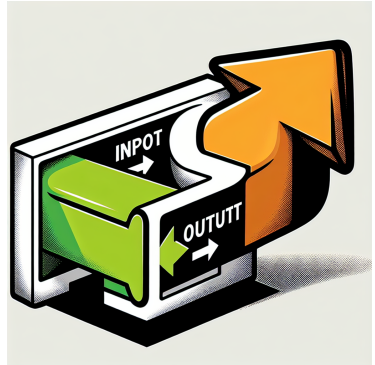
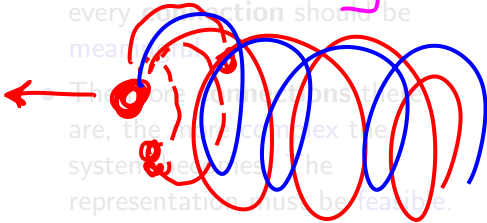


Figure: Prompt: Draw an image of a box with input and output arrows.



Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.
- Not all elements need to be connected to each other, but every **connection** should be meaningful.
- The more connections there are, the more complex the system becomes. The representation must be feasible.
- Each element must have at least one connection. Isolated elements make no sense in a system.

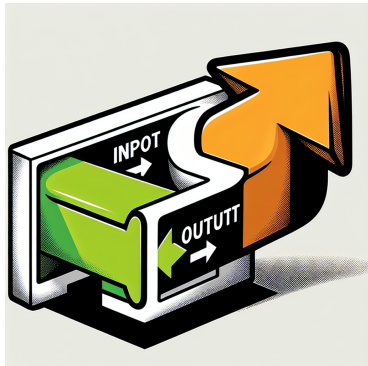


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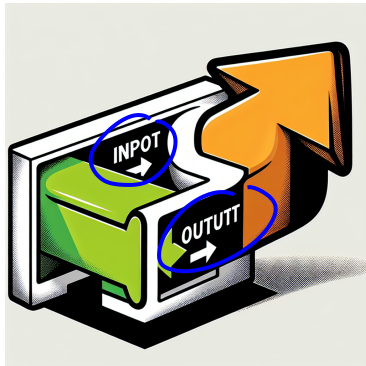


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understand
explain



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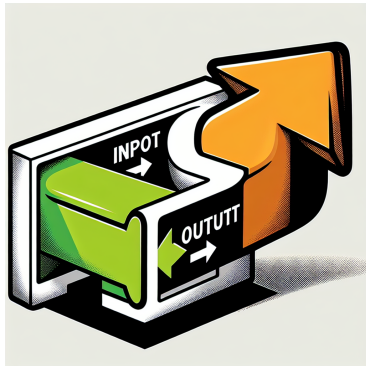


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Ing. C.A. Sierra, M.Sc. (UD FJC)



Introduction to Systems Thinking II

Dr. House

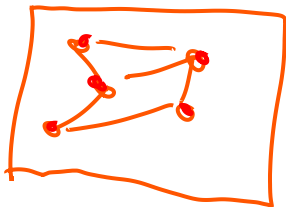


- In **systems thinking**, if you just **split** parts and forget relationships, you will **lost the full picture**.
- It is called **holistic** approach, try to see **all the picture** with all the meaning details.
- Define the **box boundaries** is sometimes tricky, as we said, not too complex, not too simple. It is like the desired universe **balance** of Thanos.

65/



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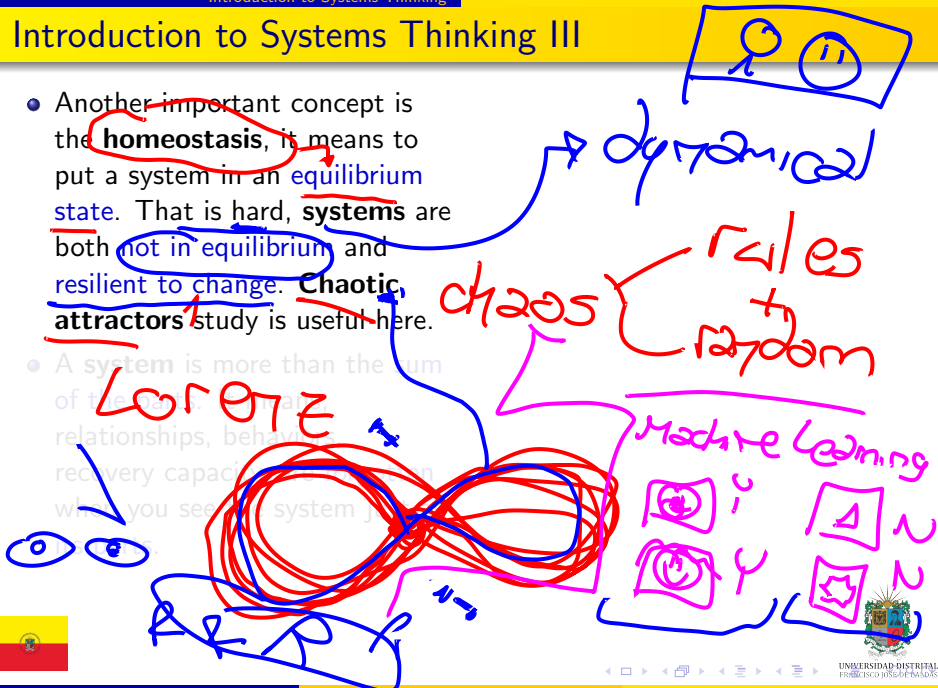
Per!



Introduction to Systems Thinking III

- Another important concept is the **homeostasis**, it means to put a system in an equilibrium state. That is hard, **systems** are both not in equilibrium and resilient to change. **Chaotic attractors** study is useful here.

- A system is more than the sum of the parts. Many relationships, behaviors, recovery capacity, etc. when you see a system just



Introduction to Systems Thinking III

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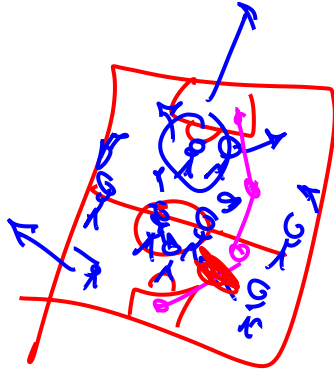
Introduction to Systems Thinking IV

- **Systems thinking** is a way to

1 understand and represent 2 problems in order to find the best possible solution.

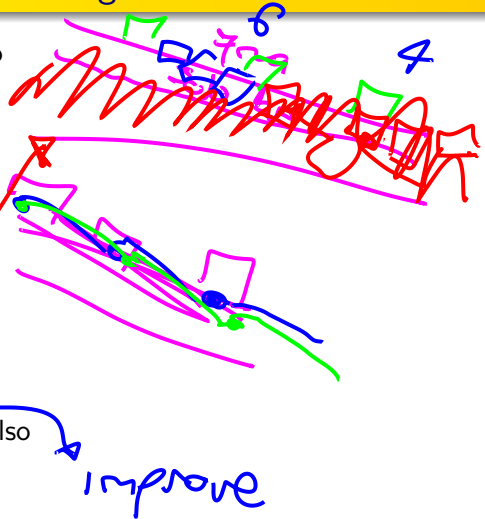
- Think in a **problem as a system** lets you understand details, involved elements, relevant information.

- Systems should be viable, auto-sostenible, provides internal feedback loops, and also looks like a whole live-entity.



Introduction to Systems Thinking IV

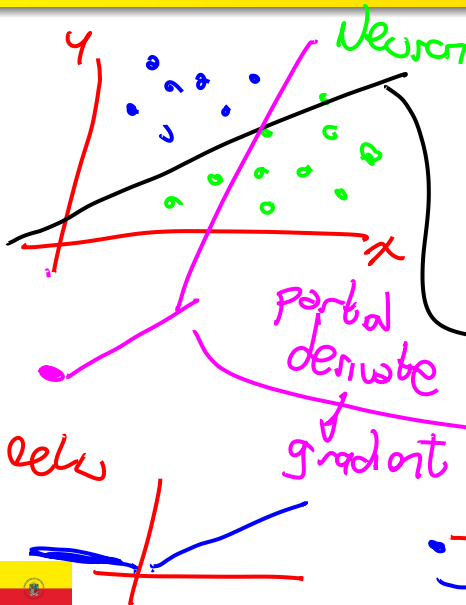
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Open AI



Introduction to Systems Thinking V



- **Computation** helps to **represent behaviors** in a mathematical way. Also, it lets to **find patterns** and information, simplify process; an example of all this is the **Artificial Intelligence**.

- **Top-Down** approach is useful when you want to see the full picture, and then split into parts. **From**
- **Bottom-Up** approach is useful when you want to see the parts and then connect them to get the full picture.



Introduction to Systems Thinking V



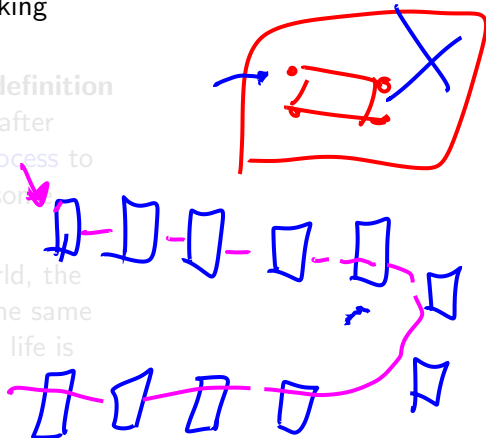
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Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to making better decisions.
- The simplest **system definition** is: given some **inputs**, after applying a designed **process** to them, you will obtain some **outputs**.
- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, real life is not deterministic.



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S.
 S.10
 S.30

6:05
 5:57

6:30

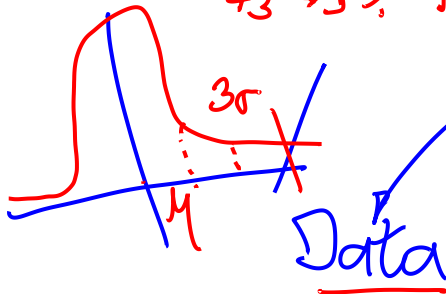


Introduction to Systems Thinking VII

5:30 — $P_1 \rightarrow 40\%$ 5

$P_2 \rightarrow 20\%$ 15

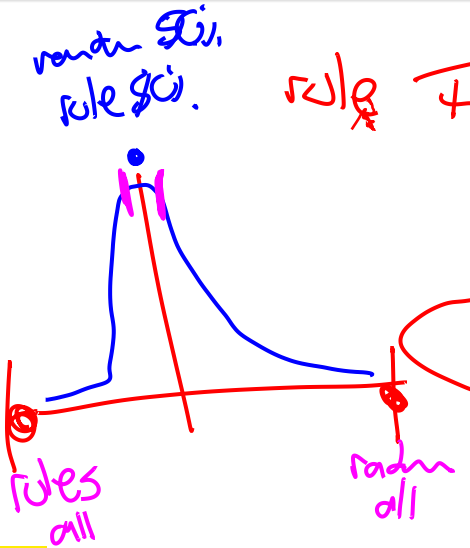
$43 \rightarrow 1\%$ 1



- Since **randomness** is normal in the real world, relying solely on deterministic processes is **dangerous**. Using **stochastic processes** is a better approach.
- **Stochastic processes** make use of **probability**, which leads to a better representation of real-world behavior.
- Here, ~~Chaos Theory~~ becomes a useful tool. To put it simply, chaos can be defined as a harmonious **balance** between rules and randomness.



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Systems Structure Draw



Inputs

name
format
level - 2015.

name
format
level - 2015.

name
format
level - 2015.

DAG

Environment



2015

Outputs

output → main

o2

o3

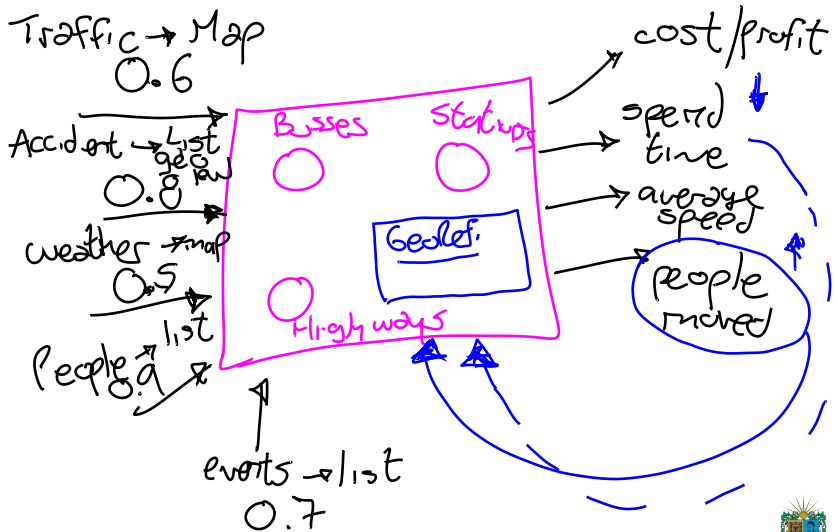
$$e/3 \cdot w1 + e/4 \cdot w2$$

cybernetics

feedback



Case of Study: Transportation System



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Systems Properties I

- **Emergence** is a property of systems that means that the **whole system** is more than the **sum of its parts**.
- **Interconnectedness** is a property of systems that means that all the **elements** are **connected** in a **meaningful** way.
- **Feedback** is a property of systems that means that the system has **internal loops** that **control** the system behavior.
- **Hierarchy** is a property of systems that means that the system has **levels of organization**.



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- **Permeability** is a property of systems that means that the system can **interact** with the **environment**.
- **Dissipative** is a property of systems that means that the system can **lose energy** and **information** to the environment.
- **Homeostasis** is a property of systems that means that the system can **maintain a stable state**.



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Basic Concepts

A **system** can be **classified** according to different **criteria** like **openness**, **adaptability**, **determinism**, and **linearity**.



Systems Classification I

- **Open systems** are systems that can **interact** with the **environment**.
- **Closed systems** are systems that **cannot interact** with the **environment**.
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Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

